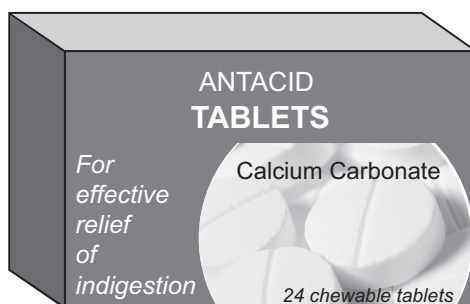


9. Indigestion is caused by excess hydrochloric acid in the stomach. The calcium carbonate in an antacid tablet neutralises the excess acid.

Calcium carbonate reacts with hydrochloric acid according to the equation below.



To determine how much calcium carbonate is present, a crushed indigestion tablet was mixed with water and titrated with dilute hydrochloric acid of concentration 1.0 mol/dm^3 . The end-point was determined using the indicator phenolphthalein. The procedure was repeated three times and the mean volume of hydrochloric acid solution needed to neutralise one tablet was found to be 15.2 cm^3 .

- (a) Calculate the number of moles of hydrochloric acid in 15.2 cm^3 of the 1.0 mol/dm^3 solution. [2]

Number of moles = mol

- (b) Calculate the number of moles of calcium carbonate in one tablet. [1]

Number of moles = mol



(c) Calculate the mass of calcium carbonate in one tablet. Give your answer in **milligrams**, **mg**. [3]

$A_r(\text{Ca}) = 40$ $A_r(\text{C}) = 12$ $A_r(\text{O}) = 16$

Mass of calcium carbonate = mg

(d) The diagram shows the labelling on the packet of indigestion tablets.

Each tablet contains:	
calcium carbonate	680 mg
magnesium carbonate	80 mg
sucrose and glucose	250 mg
flavouring	
talc	
saccharin	
sodium and magnesium stearate	

Suggest an explanation for the difference between your answer in part (c) and the information given on the packet. [1]

.....

.....

7

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1 2 3 4 5 6 7 0

		1 H Hydrogen 1				4 He Helium 2	

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE**

3410UB0-1



S19-3410UB0-1

THURSDAY, 16 MAY 2019 – MORNING**CHEMISTRY – Unit 2:****Chemical Bonding, Application of Chemical Reactions and Organic Chemistry****HIGHER TIER**

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 7(c) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	7	
3.	4	
4.	9	
5.	7	
6.	12	
7.	11	
8.	13	
9.	8	
Total	80	

3410UB01
01

MAY193410UB0101

9. (a) A student carries out a series of chemical tests on solutions of three unknown compounds, **A**, **B** and **C**. Her results are recorded in the table.

	A	B	C
Add dilute HCl	no reaction	fizzes	no reaction
Add $\text{BaCl}_2(\text{aq})$	white precipitate forms	no reaction	no reaction
Add $\text{NaOH}(\text{aq})$	green precipitate forms	pungent smelling gas given off, turns damp red litmus paper blue	no reaction
Add $\text{AgNO}_3(\text{aq})$	no reaction	no reaction	cream precipitate forms
Flame test	no colour	no colour	apple-green flame

Use the information provided to give the **chemical name** for each of the compounds. [3]

Compound **A**

Compound **B**

Compound **C**



(b) A technician wants to prepare 250 cm^3 of a 0.25 mol/dm^3 solution of lead nitrate, $\text{Pb}(\text{NO}_3)_2$.

(i) Calculate the number of moles of lead nitrate required to make the solution. [2]

Number of moles = mol

(ii) Calculate the mass of solid lead nitrate that should be dissolved to make the solution. [2]

$A_r(\text{Pb}) = 207$

$A_r(\text{O}) = 16$

$A_r(\text{N}) = 14$

Mass = g

(iii) The only electronic balance available to the technician has a precision of $\pm 0.01\text{ g}$.

Exactly how much lead nitrate should the technician weigh out to ensure that the concentration of the solution is as close as possible to 0.25 mol/dm^3 ? [1]

Mass g

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

1 H Hydrogen 1 4 He Helium 2																																			
7 Li Lithium 3		9 Be Beryllium 4																		11 B Boron 5		12 C Carbon 6		14 N Nitrogen 7		16 O Oxygen 8		19 F Fluorine 9		20 Ne Neon 10					
23 Na Sodium 11		24 Mg Magnesium 12																		27 Al Aluminium 13		28 Si Silicon 14		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17		40 Ar Argon 18					
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		128 Te Tellurium 52		127 I Iodine 53		131 Xe Xenon 54	
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																															

Key

Key

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3410UB0-1



Z22-3410UB0-1

FRIDAY, 27 MAY 2022 – MORNING

CHEMISTRY – Unit 2:
Chemical Bonding, Application of Chemical Reactions
and Organic Chemistry

HIGHER TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **11** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

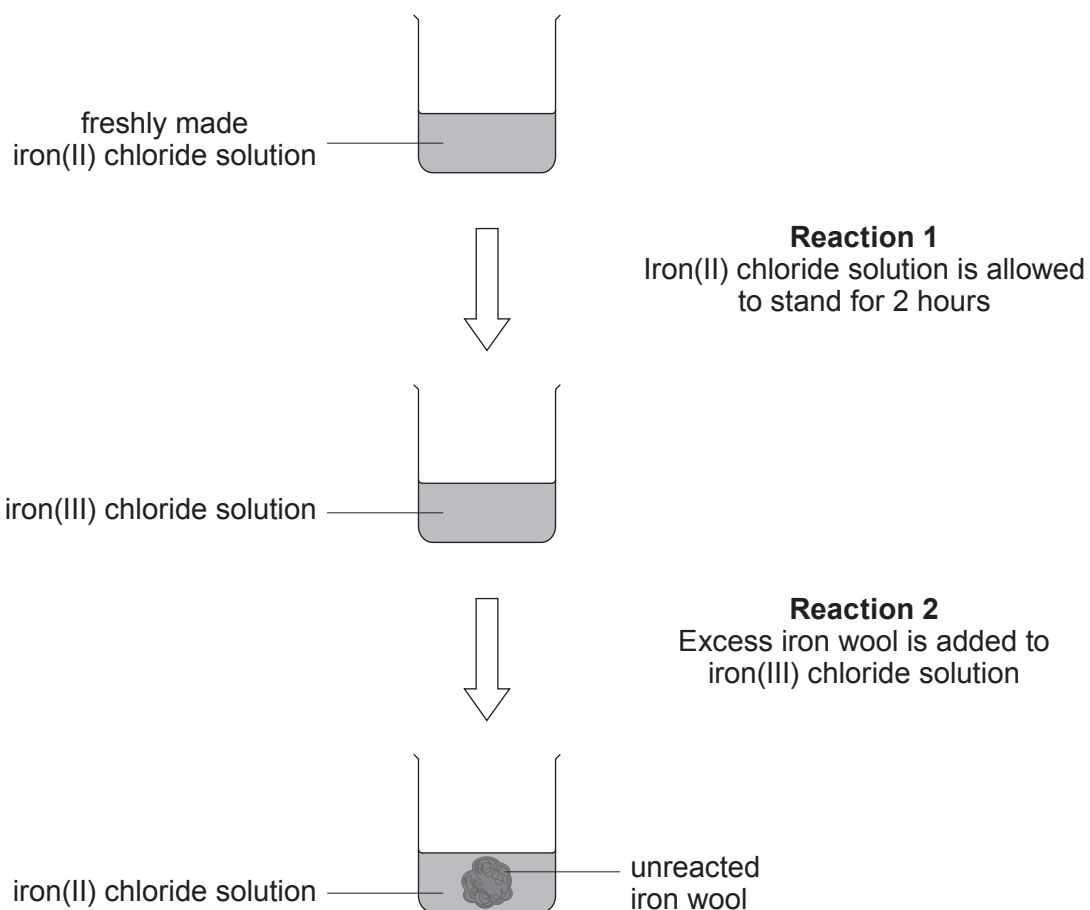
For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	6	
2.	7	
3.	7	
4.	5	
5.	9	
6.	10	
7.	5	
8.	9	
9.	7	
10.	9	
11.	6	
Total	80	

3410UB01
01

JUN223410UB0101

9. Transition metals have the ability to form ions with different charges. Iron and copper are transition metals.

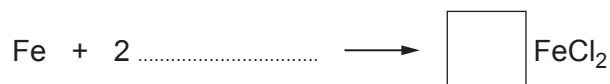
- (a) A teacher carried out a series of reactions to show that iron can form Fe^{2+} and Fe^{3+} ions.



- (i) State what you would expect to **see** during the reactions that shows iron is a transition metal. [1]

- (ii) In **reaction 2**, iron reacts with iron(III) chloride forming iron(II) chloride, FeCl₂.

- I. Complete and balance the equation for this reaction. [2]



- II. Explain the meaning of the term *oxidation* in relation to **reaction 2**. [2]

- (b) Name the reagent used to identify copper(II) ions in solution. Give the observation expected. [2]

Reagent

Observation



9. Aluminium metal is produced industrially using electrolysis, due to the relatively high stability of aluminium oxide.

The overall equation for the reaction is



- (a) Describe, in terms of electrons, the reduction and oxidation occurring during the electrolysis.

[2]

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Examiner
only

(b) (i) Aluminium ore contains 36 % aluminium oxide, Al_2O_3 .

Calculate the mass of aluminium oxide in 500 tonnes of the ore. [1]

Mass = tonnes

(ii) Calculate the mass of aluminium metal that could be produced from this mass of aluminium oxide. [3]

$A_r(\text{Al}) = 27$ $A_r(\text{O}) = 16$

Mass = tonnes

6

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group

1 2 3 4 5 6 7 0

1 H Hydrogen 1																		4 He Helium 2	
																		19 F Fluorine 9	20 Ne Neon 10
																		35.5 Cl Chlorine 17	40 Ar Argon 18
																		80 Br Bromine 35	84 Kr Krypton 36
																		127 I Iodine 53	131 Xe Xenon 54
																		210 At Astatine 85	222 Rn Radon 86
11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	27 Al Aluminium 13	28 Si Silicon 14	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	222 Rn Radon 86		
																		80 Hg Mercury 80	
																		79 Au Gold 79	
																		78 Pt Platinum 78	
																		77 Ir Iridium 77	
																		76 Os Osmium 76	
																		75 Re Rhenium 75	
																		74 W Tungsten 74	
																		73 Ta Tantalum 73	
																		72 Hf Hafnium 72	
																		57 La Lanthanum 57	
																		89 Ac Actinium 89	
45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53
86 Rb Rubidium 37	88 Sr Strontium 38	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	222 Rn Radon 86
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Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

28

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